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AN ANALYTICAL STUDY OF OPERATIONAL EFFICIENCY AND CAPACITY UTILIZATION IN TERTIARY CARE HOSPITALS

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ABSTRACT:

This study analyzes operational efficiency and capacity utilization in tertiary care hospitals, focusing on the effective use of beds, ICUs, operating rooms, diagnostic units, and OPD services. The study uses a quantitative, retrospective, and observational approach based on 12 months of primary and secondary data collected through hospital records, staff interactions, and direct observation. Key performance indicators included Bed Occupancy Rate (BOR), Average Length of Stay (ALOS), Bed Turnover Rate (BTR), ICU occupancy, OT utilization, diagnostic utilization, and OPD utilization. The findings indicate generally efficient resource utilization across most hospital departments. ALOS remained stable, ICU demand was consistently high, bed occupancy was within an appropriate range, and OPD and diagnostic services were adequately utilized. Major surgical departments demonstrated effective OT utilization, although some specialist departments showed lower utilization. The study suggests scope for improvement through better ICU surge planning, specialist OT scheduling, and enhanced utilization of advanced diagnostic resources through regular monitoring and effective planning.

Keywords: *Operational Efficiency; Capacity Utilization; Tertiary Care Hospitals; Bed Occupancy Rate; Average Length of Stay; ICU Utilization; Operating Theatre Utilization; Diagnostic Services; OPD Utilization; Healthcare Resource Management.*

1. INTRODUCTION

Tertiary care institutions that offer specialized, cutting edge medical services, like critical care, complex procedure, multidisciplinary therapy and advanced diagnostics, are basically the high point of healthcare delivery. “In practice, these organizations need a lot of resources, rely on advanced technology, and run operations that can get pretty complicated. As patient demand keeps rising and healthcare systems run into more financial and infrastructure limitations, the effective use of hospital resources has turned into big challenge for hospital administrators, healthcare managers, and even legislators.”

In hospitals, operational efficiency is usually defined as the ability as the ability of healthcare facilities to deliver high quality care in a timely, efficient, and economical way, by making the best use of what is already there. “On the other hand, capacity utilization is more about how much hospital resources – including beds, intensive care units, operating rooms, diagnostic equipment, staff, and outpatient services – are actually used compared to their available capacity.” Overall, hospital performance management leans heavily on both capacity utilization and operational efficiency, rather than only one side of the story.

When tertiary care hospitals runs in a less than smooth way, it can trigger overcrowding, longer wait times for patients, delayed admission and discharges, and a kind of chronic underutilization of costly infrastructure too. “That, in turns, usually means higher operating expenses, plus lower patient satisfaction. On the flip side, if capacity use is tuned properly, you get better clinical results along with financial wins, you boost productivity, cut down waste, and improve the way services are delivered overall.”

2. LITERATURE REVIEW OF LITERATURE

Healthcare facilities are complex services organizations, and their effectiveness seems to depend on the way infrastructure, personnel, procedures, and clinical services get coordinated, as the literature on hospital operational efficiency and capacity utilization says. “Hospital efficiency is kinda multifaceted too, so it really should be checked with operational results as well as utilization indicators, based on a substantial body of literature.”

According to Imani et al. (2022), bed occupancy, average length of stay, staff productivity, and resource allocation are some of the most important factors influencing hospital performance. Their thorough research highlighted that process design, managerial coordination, and infrastructure all have an impact on hospital efficiency.

Bed Occupancy Rate (BOR), Average Length of Stay (ALOS), and Bed Turnover Rate (BTR) are dependable measures of inpatient efficiency, according to Vaz et al. (2007), who examined bed usage indices at a tertiary

care hospital. According to their findings, hospitals with stable ALOS and optimum BOR have higher operational output.

Longer hospital stays are linked to delayed discharge, delayed diagnosis, poor care coordination, and fewer available beds, according to studies by Stone et al. (2022) and Lequertier et al. (2021). ALOS was identified in both investigations as a significant factor influencing hospital throughput and inpatient efficiency. Using the Barber–Johnson model to assess bed efficiency, Pratama et al. (2024) came to the conclusion that balancing BOR, LOS, Turnover Interval, and Bed Turnover is essential to efficient bed utilization. Their research demonstrated that inadequate bed management has a direct impact on productivity and service quality.

According to Schouten et al.'s (2023) analysis of operating room performance measures, scheduling accuracy, start-time adherence, turnover time, and case preparation all affect OT efficiency. Poor OT scheduling lowers surgical productivity and increases idle time, according to their study.

Additionally, Gupta et al. (2024) discovered that OT use varies greatly amongst departments, with specialist departments frequently exhibiting lower utilization as a result of inadequate case scheduling and unequal workload distribution.

According to Rachuba et al. (2024), integrated planning involving beds, personnel, operating rooms, and diagnostics is essential to hospital efficiency. Their review indicates that fragmented planning results in inadequate coordination and mismatched capacity.

The body of research indicates that a single factor does not determine hospital efficiency. Rather, it is a multifaceted construct that is impacted by interdepartmental cooperation, bed utilization, patient flow, ICU demand, OT scheduling, and diagnostic turnaround.

3. PROBLEM IDENTIFICATION

To provide specialized and emergency healthcare services, tertiary care hospitals works in a setting that is really fast moving and also kinda resources intensive, with many departments running at once. “Even with sophisticated infrastructure and trained staff, these hospitals often run into operational inefficiencies, mostly because coordination is not as tight as it should be, workload is not balanced evenly, and available resources are used in a way that is not that efficient. So, to understand where the performance drops happen and to improve overall hospital productivity, those problems have to be spotted first.”

The big issue in tertiary care facilities is basically the mismatch between what patients demand and what services capacity is actually there. “Because of this imbalance, hospital resources can end up being used too

much in some places and then in other areas they underused. You might see certain departments staying underutilized even through they cost a lot to maintain while other departments gets hit with heavy patient volumes, longer queues, and delays that just keep stacking up." This kind of operational mismatch affects the quality of care, pushes cost upward and overall reduces efficiency, in a pretty direct way.

Key Issues Found

- Delays in bed rotation and high bed occupancy
- Extended wait times for patients in emergency rooms and outpatient departments
- Bottlenecks in admission and delayed discharge
- Limited surge flexibility and high ICU occupancy
- Inconsistent use of operating rooms across departments
- Postponements in reporting and diagnostic tests
- Inadequate use of cutting-edge diagnostic resources (CT/MRI)
- Unbalanced workloads and resource exhaustion among employees
- Inadequate departmental cooperation
- Restricted usage of systems for real-time monitoring

Both operational effectiveness and patient outcomes are impacted by these issues. Overcrowding, service delays, ineffective workflow, and higher operating costs are the results.

4. METHODOLOGY

To investigate operational effectiveness and capacity utilization in tertiary care institutions, the current study used a descriptive and analytical technique. This approach was used to evaluate how well hospital resources are used and how important hospital indicators can be used to quantify operational performance.

4.1 Research Design

A descriptive and analytical research design was used for the investigation.

• Hospital operations and resource usage patterns were explained using a descriptive design.
• Hospital performance was interpreted and operational inefficiencies were found using analytical design.

4.2 Nature of Study

The study was observational, quantitative, and retrospective. • Because it concentrated on quantifiable hospital signs, it was quantitative. • Quantitative since it concentrated on hospital metrics that could be measured. • Observational since hospital procedures were examined in their natural environment. • Observational, as

hospital procedures were examined in their natural environment. • Retrospective since the study made use of previously released data and records.

4.3 3 Sources of Data

Secondary data served as the study's primary source.

Among the data sources were:

- Articles of research
- Research on hospital administration
- Reviews that are systematic
- Reports on operations
- Studies on hospital utilization

The data needed to analyze hospital efficiency and capacity usage came from these sources.

4.4 Variables of Study

Important hospital performance metrics utilized in the study included:

- BOR, or bed occupancy rate
- ALOS, or average length of stay
- BTR, or bed turnover rate
- BTI, or Bed Turnover Interval
- Rate of ICU Occupancy
- Rate of OT Utilization
- Rate of Diagnostic Utilization
- Rate of OPD Utilization

The operational effectiveness of several hospital service areas was assessed using these characteristics.

4.5 Data Analysis Method

Simple analytical methods were used to examine the data, including:

- Analysis of percentages
- Comparative evaluation
- Analysis of trends
- Ratio analysis

These techniques aided in determining operational gaps and assessing usage levels.

4.6 Presentation of Data

The data analysis was displayed using:

- Tables
- Comparisons by department
- Interpretations of trends

5. RESULTS AND DISCUSSION

The findings of the study indicate that the selected tertiary care hospital demonstrated generally satisfactory operational efficiency and effective utilization of major hospital resources. However, the results also identified specific operational gaps requiring improvement.

5.1 Bed Occupancy Rate (BOR)

Throughout the study period, the Bed Occupancy Rate varied from 82% to 86%, demonstrating effective use of inpatient bed resources. In tertiary care institutions, this range is ideal since it shows high bed usage without undue overcrowding. Peak occupancy in some months, however, indicated short-term strain on bed availability and less room for emergency admissions.

Table 5.1 Rate of Bed Occupancy (BOR)

Month	Available Bed Days	Occupied Bed Days	BOR (%)
January	3000	2460	82
February	2800	2296	82
March	3000	2580	86
April	3000	2490	83
May	3000	2550	85
June	3000	2520	84

5.2 Average Length of Stay (ALOS)

The average length of stay stayed steady between 5.6 and 5.8 days, indicating successful discharge planning, constant treatment duration, and efficient inpatient care. A stable ALOS shows that patient throughput was maintained without undue treatment or discharge delays.

Table 5.2 Average Length of Stay (ALOS)

Month	Inpatient Days	Discharges	ALOS (Days)
January	2460	430	5.7
February	2296	410	5.6
March	2580	450	5.7
April	2490	440	5.6
May	2550	445	5.7
June	2520	438	5.8

5.3 Bed Turnover Rate (BTR) and Bed Turnover Interval (BTI)

Four to five patients used each bed each month, according to the Bed Turnover Rate, which varied from 4.1 to 4.5. This indicates effective inpatient throughput and healthy bed productivity. The Bed Turnover Interval stayed near one day, suggesting that there was little idle bed time and that beds were promptly reassigned upon discharge.

Table 5.3 Bed Turnover Rate (BTR)

Month	Discharges	Beds	BTR
January	430	100	4.3
February	410	100	4.1
March	450	100	4.5
April	440	100	4.4
May	445	100	4.5

Month	Discharges	Beds	BTR
June	438	100	4.4

5.4 ICU Ocuupancy Rate

ICU occupancy remained high, ranging from 83% to 90%, suggesting that critical care resources are being used effectively. Occupancy levels close to 90% may limit surge capacity and present difficulties during emergencies or abrupt increases in the critical patient load, even though this indicates effective ICU use.

5.5 Operation Theatre Utilization

Major departments like General Surgery, Orthopedics, and Gynecology had effective OT usage rates, ranging from 80% to 83%. Specialty departments like ENT, on the other hand, had lower usage (68%), which suggests that costly surgical infrastructure is underutilized and that improved scheduling is required.

5.6 Diagnostic Utilization

The utilization of diagnostic services was satisfactory, with pathology showing the highest utilization (88%), followed by radiology and CT (85%). The use of MRI was relatively lower (80%), indicating potential for better scheduling and better use of cutting-edge diagnostic facilities.

5.7 OPD Utilization

Strong outpatient demand and effective use of OPD service capacity are demonstrated by the high OPD utilization, which ranged from 84% to 88%. However, if patient flow is not properly controlled, persistently high OPD use may result in congestion, longer wait times, and more staff labor.

5.8 Discussion

Overall, the hospital's operating efficiency in all of its main service areas was satisfactory. “Overall, there was some efficiency in how beds were used, in patient throughput, ICU use, diagnostics, and out patient services.” Still, the high ICU occupancy, the unused specialist OT capacity, and the lower MRI utilization, seems to point, to the need or better resource efficiency and a more balanced capacity.

6. CONCLUSION & SCOPE OF FURTHER WORK

CONCLUSION

The chosen tertiary care hospital did manage in general a pretty satisfactory operational efficiency and, even more, useful capacity utilization across the big departments as per the study findings. “You could see stable

operational performance when the utilization of key hospital resources - beds, ICU services, diagnostic facilities, and OPD capacity – stayed within the acceptable efficiency limits."

Overall, the study suggests that operational efficiency in tertiary care hospitals is not only tied to having higher infrastructure volume, it depends a lot on how well resources are coordinated, watched, and governed. "For that reason, continuous performance monitoring and better scheduling systems matter, as well as improved interdepartmental coordination and data driven planning, all of which are essential if the hospital wants to sustain and also keep raising its operational efficiency."

SCOPE OF FURTHER WORK

There are still a few areas that could use more research, even if the study gives some insightful information about how well hospitals run operationally. "In other words to evaluate performance and sort out the best practices, future research might also widen the scope by doing comparative analysis across multiple tertiary care facilities." Some deeper insights about specific service bottlenecks can be pulled from department focused operational studies, like emergency efficiency, ICU workflow, OT scheduling, and pharmacy logistics.

Financial efficiency metrics such as cost per occupied bed, cost per patient treated, and department-specific cost benefit evaluation might also show up in later research, if needed. "Operational studies in hospital can get stronger by adding patient flow mapping, waiting time scrutiny, and some patient satisfaction indicators too."

In coming work, investigators may use more advanced methods like queueing theory, simulation modeling, lean management, data envelopment analysis, and predictive analytics to help with capacity forecasting and process optimization.

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